

Lab 02 – MATH 240 – Computational Statistics

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Abstract

In this lab we practiced using loops and different functions to manipulate strings.

Keywords: for loops; stringr;

1 Introduction

The purpose of this lab was to use for loops to extract song names, album names, and artist names from a music folder in order to eventually compile a .bat file to run a program to get data from each song. The goal for this lab and the next 2 labs is to collect data from songs from two different artists and use the data to determine which artist contributed the most to their collaborative song.

2 Methods

The data used for this lab is a directory called Music with two sub-directories each with multiple subsequent sub-directories, each containing two .wav files.

2.1 Isolate Artist, Album, and Track

1. I created a vector of all the directories in the music folder then used `str_count()` to separate the sub-directories from main directories, which will help to obtain the files in each folder and later to isolate the artist and album names. Wickham (2023)
2. Next, I created an empty vector for the file names and used a for loop to list the files from each sub-directory and store them in the vector called files. This will also aid me in isolating the track names later.
3. Next, using `paste()`, and a for loop, I created a vector for the file paths by combining the “subdirectories” and “files” vectors in a clever way so that each file path was correct.
4. Now we have a vector that has the artist name, album name, and track name and can isolate each rather easily using `str_split_i()` and `str_sub()` in for loops.

2.2 Create the .bat File

To create the .bat file we needed to automate a process that called

```
streaming_extractor_music.exe "[track name].wav"  
"[artist name]-[album name]-[track name].json."  
for each song. streaming_extractor_music.exe is constant  
for each song, and we already have "[track name].wav" in  
the form of the “files” vector so the only step left before cre-  
ating the .bat file was to put together artist, album, track,  
and the string “.json”. To do this, I used paste() combined  
with a for loop and stored it in the “output” vector.
```

Finally, I used a for loop and `paste()` to combine the three things needed and created a vector with each line of code that we want in the .bat file called code.to.process. Then wrote the lines into batfile.txt.

2.3 Extract Data from a .json File

First I loaded the json file into an object in R using `fromJSON()`. Then was just able to look through the object using `View()` and then assign each piece of data to a vector by calling the value wanted from the “json” object. Ooms (2014)

For example,
`average_loudness = json$lowlevel$average_loudness`

3 Results

The primary result of this lab was the “batfile.txt” could now be used to run the code to extract data from the songs that we want. This will help us in the next few labs where we get the data and use it to determine which artist had a bigger part in their collaboration song.

The other result of this lab was the data from “The Front Bottoms-Talon Of The Hawk-Au Revoir (Adios).json”.

4 Discussion

No evidence or results were discovered yet, as this lab was just a set up for the next few labs.

Bibliography:

References

Ooms, J. (2014). The jsonlite package: A practical and consistent mapping between json data and r objects. *arXiv:1403.2805 [stat.CO]*.

Wickham, H. (2023). *stringr: Simple, Consistent Wrappers for Common String Operations*. R package version 1.5.1.